

Algebra 1 Honors Unit 14 Assessment Guide

General Information	Many of the benchmarks in the unit are supporting benchmarks. In other words, they are not explicitly taught but support the learning of the student. Much of the learning in the unit centers on MA.912.DP.3.1. Students are organizing categorical bivariate data into a frequency table. Students learn to consider possible associations which is further explored within the Honors benchmarks (MA.912.DP.3.2 and MA.912.DP.3.3). In the lessons, students learn how segmented bar graphs may be helpful in visualizing associations (MA.912.DP.1.1). Students interpret the data using the given information and organize the frequencies into two-way frequency tables. Honors students extend their learning to relative frequencies and interpreting frequencies. Teachers of honors students may wish to know that the benchmark that is assessed on the statewide end-of-course assessment does not require students to use relative frequencies. The draft test item specifications state that "Items will not require the student to find/interpret relative frequencies." The content of this unit, although new to the student, does not place a high cognitive demand on the student. Teachers may be tempted to skip teaching this content and replace it with reviews for the statewide end-of-course exam. Because most of the content is less demanding, teachers are encouraged to build student understanding to better equip students for success. Teachers may wish to consider that the information the teacher learned from the unit 13 assessment can be utilized to add in spiral learning to support students in areas of weakness while students have a less cognitively demanding unit.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1			Benchmark(s)	MA.912.DP.3.2
Do you like it or dislike it when strangers start conversations with you?				Recommended Scoring (18 Total Points)	Consider giving students 2 points for each value in the table.
	Like <u>It</u>	Dislike It	Total		
Male	130	212	372		
Female	169	234	403		
Total	299	476	775		
Question	2a			Benchmark(s)	MA.912.DP.3.1, MA.912.DP.3.3, MA.912.DP.1.2
	Flycatcher	Hummingbird	Total	Recommended Scoring (18 Total Points)	Consider giving students 2 points for each value in the table.
Cacao Farm	40.51%	24.05%	64.56%		
Rainforest	12.03%	23.42%	35.44%		
Total	52.53%	47.47%	100.00%		
Question	2b				
40.51% of birds found were Flycatchers on the Cacao Farm				Recommended Scoring (3 Total Points)	Consider no partial credit.
Question	2c			Benchmark(s)	MA.912.DP.3.3
35.44% of birds were found in the rainforest				Recommended Scoring (3 Total Points)	Consider no partial credit.

Question		3a				Benchmark(s)	MA.912.DP.3.1		
						Recommended Scoring (40 Total Points)	Consider giving students 2 points for each missing value in each table.		
Freshmen				Sophomores					
	Do Get <u>An</u> Allowance	Do Not Get <u>An</u> Allowance	Total		Do Get <u>An</u> Allowance			Do Not Get <u>An</u> Allowance	Total
Do Chores	96	25	121	Do Chores	131			62	193
Do Not Do Chores	60	121	181	Do Not Do Chores	18			58	76
Total	156	146	302	Total	149			120	269
Juniors				Seniors					
	Do Get <u>An</u> Allowance	Do Not Get <u>An</u> Allowance	Total		Do Get <u>An</u> Allowance			Do Not Get <u>An</u> Allowance	Total
Do Chores	101	72	173	Do Chores	105			84	189
Do Not Do Chores	71	23	94	Do Not Do Chores	21			42	63
Total	172	95	267	Total	126	126	252		
Question		3b				Benchmark(s)	MA.912.DP.3.3		
		23.68%				Recommended Scoring (2 Total Points)	Consider giving students 2 points for each missing value in the table.		
Question		3c				Benchmark(s)	MA.912.DP.3.3		
		82.88%				Recommended Scoring (2 Total Points)	Consider giving students 2 points for each missing value in the table.		

Question	4a			Benchmark(s)	MA.912.DP.3.1																
<table><tr><th></th><th>Has a Job</th><th>Does Not Have a Job</th><th>Total</th></tr><tr><th>Has Driver’s License</th><td>67.5%</td><td>22.5%</td><td>90%</td></tr><tr><th>Does Not Have a Driver’s License</th><td>2%</td><td>8%</td><td>10%</td></tr><tr><th>Total</th><td>69.5%</td><td>30.5%</td><td>100%</td></tr></table>					Has a Job	Does Not Have a Job	Total	Has Driver’s License	67.5%	22.5%	90%	Does Not Have a Driver’s License	2%	8%	10%	Total	69.5%	30.5%	100%	Recommended Scoring (10 Total Points)	Consider giving students 2 points for each missing value in the table.
	Has a Job	Does Not Have a Job	Total																		
Has Driver’s License	67.5%	22.5%	90%																		
Does Not Have a Driver’s License	2%	8%	10%																		
Total	69.5%	30.5%	100%																		
Question	4b			Benchmark(s)	MA.912.DP.3.1, MA.912.DP.1.2, MA.912.DP.1.3																
About $\frac{3}{4}$ or the seniors surveyed that have a driver’s license have a job, whereas only about $\frac{1}{5}$ of those who don’t have a license have a job. This shows that there is an association between having a driver’s license and having a job.				Recommended Scoring (4 Total Points)	Consider no partial credit.																